

Session Stat+



Statistical
approach

JJumbo



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Introduction

The **SessionStat+** indicator is a powerful and versatile tool designed to empower traders with precise insights into market price dynamics across multiple timeframes and custom sessions. By leveraging statistical analysis of historical price movements, **SessionStat+** identifies key price levels, including expected highs, lows, and maximum/minimum distributions, enabling traders to anticipate market behavior with confidence.

Whether you're trading on 15-minute, hourly, daily, weekly, or custom-defined sessions, **SessionStat+** calculates critical metrics using both simple and weighted averages, as well as medians. These calculations are visualized through intuitive lines, boxes, and projections, providing a clear roadmap of support, resistance, and breakout zones. With customizable settings for data lookback periods, timezones, and visual styles, **SessionStat+** adapts to any trading strategy, from scalping to swing trading.

Key features include:

- **Multi-Timeframe Analysis:** Analyze price levels across standard timeframes or up to four custom sessions.
- **Statistical Projections:** Forecast potential price targets with upper and lower expansion zones based on historical data.
- **Minimum Average levels:** Detects potential swings areas that fall short from highs and lows.
- **Visual Clarity:** Display levels with customizable lines, shaded boxes, and midpoint lines for enhanced chart readability.
- **Flexible Configuration:** Tailor session times, averaging methods, and visual preferences to suit your trading style.

Whether you're a day trader, swing trader, or market analyst, **SessionStat+** equips you with the data-driven edge needed to make informed decisions in dynamic markets.

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Indicator instructions

SessionStat+ [JJumbo]

Inputs

Style

Visibility

PERIOD SETTINGS

Type

Daily

Lookback

60

Average Type

Simple

Timezone

America/New

CUSTOM SESSION SETTINGS

Extension Type

Extend L...

☒

09:00

–

12:00

☐

00:00

–

16:00

☐

02:00

–

12:00

☐

12:00

–

14:00

☒

Avg/Med Box?

Line Color

Line Style

Solid

Label Color

Label Size

tiny

M15

H1

H4

Daily

Weekly

☒

Show Midpoint Line?

Midpoint Line Style

Dotted

Midpoint Line Color

Daily

15 Minute

1 Hour

4 Hour

Daily

Weekly

Custom Session

Period - Custom session settings

- Fixed timeframe selection
- Custom Lookback Data
- Custom timeframe selection with up to 4 sessions
- 2 types of average calculation
- Range open and close levels
- Customizable lines, boxes, labels
- Data table displaying average range of selected session (Min, Max, Expansion)

PROJECTION SETTINGS

☒ Show Projections?

Upper Expansion

Lower Expansion

Line Style +/- Expansion

Projection High Color

Projection Low Color

MINIMUM AVERAGE LEVELS

☐ Show Min Avg Levels?

Line Style

Line Width

Line Color

☐ Show Min Avg Labels?

☒ Show Min Avg Shading?

Shading Color

Projections settings

- Projections based on the statistical levels
- Customisable projections (high - low)

Minimum Average levels

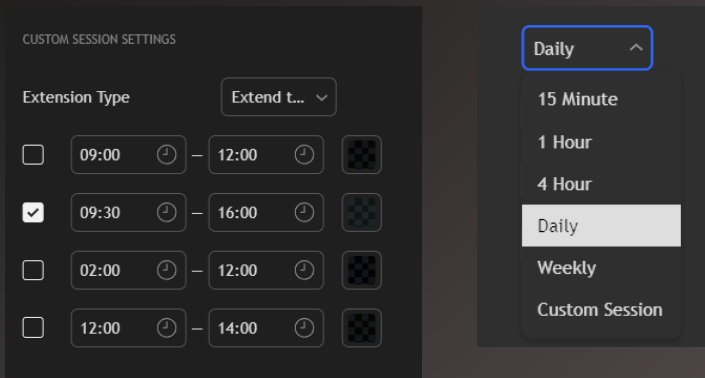
- Minimum Average range display of selected sessions
- Customizable lines, labels, boxes



Functionality

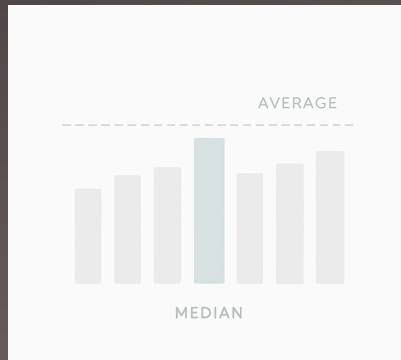
- **Flexible Period Analysis:**

- The indicator allows users to select a time period for analysis: 15-minute, 1-hour, 4-hour, daily, weekly, or custom sessions.
- For custom sessions, users can define up to four distinct sessions with specific start and end times, each with customizable colors for visualization.



- **Level Calculations:**

- **Average and Median Levels:** It calculates the average and median of the Highs and Lows over the lookback period chosen.
- **Minimum average Levels:** It computes additional levels based on the minimum average of the session, which reflects price movements of shorter swings.
- **Projections:** Optional custom upper and lower projections are calculated by extending the average range (difference between upper and lower levels).



Accuracy & Data Detection

The levels in the indicator may differ when switching to a timeframe below 2 minutes by design, to gather additional accuracy in the levels. Consider using the indicator above 2-3 minute time-frame (unless using the 1 hour or 15 minute non-custom chosen session)

Bar-Based Calculations and Session Detection:

- The indicator uses `bar_index` and time-based conditions to detect the start and end of periods or sessions. On lower timeframes, there are more bars per session, which can lead to more frequent updates or slight variations in how session boundaries are detected due to timestamp granularity.
- For example, a 1-minute chart has 5 times more bars than a 5-minute chart, so the exact bar where a session starts or ends might differ slightly, affecting when the indicator initializes or updates session data.

Practical Implications:

- **Visual Differences:** On timeframes below 5 minutes, you might notice levels appearing at slightly different prices or extending to different bars due to the finer granularity of data and potential misalignment in session detection.
- **Performance Considerations:** Lower timeframes generate more bars, increasing computational load and potentially causing the indicator to process data differently, especially for custom sessions with frequent updates.

Custom Sessions time limitations

The Custom sessions feature are designed to work in current day time stamps only (Midnight to Midnight), when selecting cross-day times (e.g. 11pm - 5am) the indicator will automatically stop plotting.

Single-Day Timestamp Logic:

- The indicator sets session start and end times using the current day's date. For example, when you define a session like 9:00 AM–12:00 PM, it assumes both times are within the same day (e.g., today). If you try to set a session that crosses midnight (e.g., 11:00 PM yesterday to 2:00 AM today), the indicator doesn't account for the day change because it uses the current day's date for all timestamps.

If you want to set a custom session starting at 6:00 PM and ending at or near midnight (e.g., for the Asia session such as 8:00 PM to 11:59 PM), there are specific settings you need to use due to how the indicator handles session timing and resets.

Extension Type

Extend ... ^

Extend Until Session End

Extend to End of Day

Daily Reset at Midnight (00:00):

- The indicator ties session timestamps to the current day's date. At midnight (00:00 in the specified timezone, default America/New_York), the date changes, and the indicator treats this as a new day, resetting session data. If a session's end time is set to 00:00 or beyond, it may be interpreted as the next day, causing the session to fail or not plot correctly.
- To avoid this, you should set the session end time to 11:59 PM (23:59), the last minute of the current day and set **"Extension Type"** to **"Extend Until Session End"**

Layouts

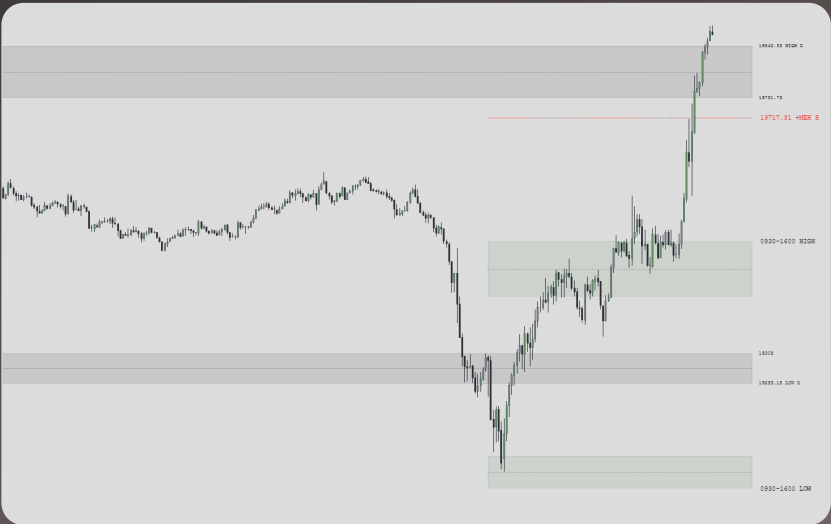
09:00 - 12:00 Session



09:30 - 16:00 RTH Session



Daily + RTH Session



Key Session Times

9:00 AM–12:00 PM: New York AM Session (High and Low)

What It Represents

- **Session Time:** 9:00 AM to 12:00 PM New York time.
- **Purpose:** This session captures the New York AM session, which represents the morning trading activity in the U.S. financial markets, particularly for Index Futures
- **High and Low:** The indicator calculates the highest price (high) and lowest price (low) during this 3-hour window (9:00 AM–12:00 PM) on the current day. These levels are significant as they reflect the price range during the initial hours of active trading in New York, often a period of high volatility and liquidity.

9:30 AM–4:00 PM: Regular Trading Hours (RTH) Session (High and Low of Day)

What It Represents

- **Session Time:** 9:30 AM to 4:00 PM New York time.
- **Purpose:** This session corresponds to the Regular Trading Hours (RTH) for major U.S. stock exchanges, such as the NYSE and NASDAQ, when most trading volume occurs. It captures the high and low of the trading day, representing the full range of price action during official market hours.
- **High and Low:** The indicator calculates the highest price (high) and lowest price (low) from 9:30 AM to 4:00 PM, effectively covering the entire day's primary trading activity. These are considered the daily high and low because RTH encompasses the most significant trading period.

Levels Failure Potential

When and Why Levels Might Fail

Levels generated by the indicator may fail to act as reliable support/resistance or be breached unpredictably in the following scenarios:

Choppy Market Conditions:

- Why It Happens:
 - In choppy markets, prices oscillate rapidly within a narrow range, often triggered by lack of directional momentum.
 - The indicator's levels are based on average historical ranges, which expect a certain degree of directional movement. Choppy conditions produce erratic, non-directional price action, which may misalign with these averages.
- Impact on Levels:
 - High/Low Levels: The projected high/low may be repeatedly tested and breached in both directions, failing to hold as support/resistance.
 - Projections: Extended projections may be too far from the choppy price action, rendering them irrelevant.
 - Boxes: Boxes between average and median levels may be too wide or narrow, as choppy prices don't respect historical ranges.

Low Volatility Environments:

- Why It Happens:
 - In low-volatility periods, price ranges are much smaller than historical averages due to reduced trading activity or market interest. For instance, during a holiday or weekend, trading volume may drop, leading to tight price ranges.
 - The indicator's levels are based on past sessions with normal volatility, which may overestimate the expected high/low in a low-volatility market.
- Impact on Levels:
 - High/Low Levels: Projected levels are too far from the actual price action. For example, if the historical average expansion is 10 points, but today's range is only 2 points, the high level at open + 10 is unreachable.
 - Projections: Projections are even further out, making them irrelevant.
 - Boxes: Boxes may appear overly wide, as the actual price range is much smaller than the historical average range.

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Stay tuned for future updates and releases as well video presentations and trade breakdowns through my X account or Youtube



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